where $(m_i, \overline{m}_j)_{(i,j)\in(E_0\times E_0)} \in (\mathbb{R}^d\times\mathbb{R}^d)^{E_0\times E_1}$ and $(\sigma_i, \overline{\sigma}_j)_{(i,j)\in(E_0\times E_0)} \in (\mathcal{S}_d^+\times\mathcal{S}_d^+)^{E_0\times E_1}$ are given parameters. Combining Proposition 5.4 with Proposition 5.3 we readily obtain the following result.

Proposition 5.5. *The Gaussian mixture model defined in (5.14) with the Gaussian log-densities of (5.15) defined on $\mathbb{X} = \mathbb{Y} = \mathbb{R}^d$ satisfies Condition $\mathcal{H}_\delta(U, V)$ for some $\delta \in]0, 1[$ provided that*

$$\beta'\tau^{-1}\beta < (\delta \wedge (1-\delta))\ \sigma_i^{-1} \quad \text{and} \quad \tau^{-1} < (\delta \wedge (1-\delta))\ \overline{\sigma}_j^{-1} \tag{5.16}$$

for almost every $(i, j) \in (E_0 \times E_1)$.

Let $(X_i)_{1\leqslant i\leqslant N}$ be N independent and identically distributed samples from some (arbitrary) data distribution $\eta(dx)$ on $\mathbb{X} = \mathbb{R}^d$. The Gaussian-kernel estimation model defined by

$$\lambda_U(dx) := \frac{1}{N} \sum_{1\leqslant i\leqslant N} g_\sigma(x - X_i)\ dx \tag{5.17}$$

is a well known Gaussian mixture approximation of the distribution $\eta(dx)$ (as the number of samples $N \to \infty$). The covariance matrix σ is a bandwidth parameter that controls the amount and orientation of the smoothing. Note that (5.17) coincides with (5.14) when $(m_i, \sigma_i) = (X_i, \sigma)$ with the counting measure $\xi_0(i) = 1/N$ on the index set $E_0 = \{1, \ldots, N\}$. For a more detailed discussion on these non parametric statistical estimation techniques we refer to the pioneering articles [86, 92] as well as the books [16, 96, 107].

The Gaussian mixture distribution (5.17) can be used to estimate complex and unknown data distributions arising in generative models, see for instance [36] and references therein.

Remark 5.6. *Schrödinger bridges and the Sinkhorn algorithm are often used in practice to solve high dimensional optimal transport problems. In our context, choosing $\tau = tI$ and $(\alpha, \beta) = (0, I)$ in (5.7), the Schrödinger bridge in the limit $t \to 0$ of vanishing regularization solves the Monge-Kantorovich problem and converge towards the optimal 2-Wasserstein coupling. In this case, $\mathcal{S}_0 = Q$ coincides with the heat kernel on a time horizon t (see also Remark 2.3 in [2]), namely,*

$$\mathcal{Q}(x, dy) = (2\pi t)^{-d/2}\ \exp\left(-\frac{1}{2t}\ \|y - x\|^2\right)\ dy.$$

with the Euclidian norm $\|z\|^2 = z'z$. In this context, the l.h.s. condition in (5.16) applied to (5.17) becomes

$$\frac{1}{t}\ I < (\delta \wedge (1-\delta))\ \sigma^{-1}. \tag{5.18}$$

In many cases, the Markov transition $\mathcal{Q}(x, dy)$ of interest is generated by a stochastic diffusion between time 0 and a time horizon t –see, e.g., [78, 101, 69]. In our context, if we choose $(\alpha, \beta) = (0, e^{tA})$ and $\tau = \tau_t := \int_0^t e^{sA}e^{sA'}ds$ for some Hurwitz matrix A and some $t > 0$, $\mathcal{Q}(x, dy)$ coincides with the transition semigroup of the Ornstein-Uhlenbeck diffusion (see Remark 2.3 in [2]). In this case, the l.h.s. condition in (5.16) applied to (5.17) is met for any $t \geqslant t_0$ provided that

$$e^{tA'}\tau_{t_0}^{-1}e^{tA} < (\delta \wedge (1-\delta))\ \sigma^{-1}. \tag{5.19}$$

Both transitions gradually transform the data distribution λ_U into a simpler distribution $\lambda_U\mathcal{Q}$ by injecting noise into the data. Also note that (5.18) and (5.19) are met when the time horizon is chosen sufficiently large.

The choice of the prior distribution ν_V is rather flexible. For instance, we can choose a Gaussian distribution, say $\nu_{0,\overline{\sigma}}$, satisfying the r.h.s. condition in (5.13). Starting from such Gaussian random states, the backward Sinkhorn transitions (indexed by odd indices) eventually generate samples back into the initial distribution by slowly removing the noise distortion. Conditions

(5.18) and (5.19) reflect the fact that the forward noise injection as well as the reverse denoising procedure in diffusion-based generative models often require very large time horizons.

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Appendix

This appendix compiles technical material and supplementary results that bolster the paper's central arguments. Appendix A focuses on Gaussian Sinkhorn algorithms. This material is taken from [2]. We review some properties of the Riccati matrix differential equations and the random maps associated with Sinkhorn transitions. We also provide equivalent formulations of the drift-minorization conditions for the Lyapunov functions presented in Proposition 5.2. Appendix B is devoted to basic properties of Hilbert's projective metric. Appendix C recalls several analytical tools related to weighted Banach spaces, operator norms, and contraction coefficients that are used throughout the stability analysis. For a more thorough discussion on weighted norms and contraction coefficients, we refer the reader to the articles [40, 43, 41, 39] as well as the book [44].

Appendix D reviews Sinkhorn potentials. The first part, Appendix D.1, presents some commutation formulae. Building on this framework, several quantitative estimates are provided in Appendix D.2. Section D.3 is dedicated to Lyapunov inequalities and minorization conditions. The Proof of Theorem 4.8 is provided in Appendix D.4 and Appendix E contains further technical arguments.

Appendix A. The Gaussian Sinkhorn algorithm

The Frobenius matrix norm of a given matrix $v \in \mathbb{R}^{d_1 \times d_2}$ is defined by $\|v\|_F^2 = \mathrm{Tr}(v'v)$, with the trace operator $\mathrm{Tr}(\cdot)$ and v' the transpose of the matrix v. Note that, for $x \in \mathbb{R}^{d\times 1} = \mathbb{R}^d$, the Frobenius norm $\|x\|_F = \sqrt{x'x}$ coincides with the Euclidean norm. The spectral norm is defined by $\|v\|_2 = \sqrt{\lambda_{\mathsf{max}}(v'v)}$.

Riccati difference equations. Let us associate with some $\varpi \in \mathcal{S}_d^+$ the increasing map Ricc_ϖ from $\mathcal{S}_d^0$ into $\mathcal{S}_d^+$ defined by

$$\mathrm{Ricc}_\varpi : \begin{array}{ccc} \mathcal{S}_d^0 & \mapsto & \mathcal{S}_d^+ \\ v & \rightsquigarrow & \mathrm{Ricc}_\varpi(v) := (I + (\varpi + v)^{-1})^{-1} \end{array} . \tag{A.1}$$

If we define the matrices

$$\varpi^{-1} := \gamma\,\gamma' \quad \text{and} \quad \overline{\varpi}^{-1} := \gamma'\gamma \in \mathcal{S}_d^+, \quad \text{with} \quad \gamma := \overline{\sigma}^{1/2}\,\tau^{-1}\beta\,\sigma^{1/2},$$

and $\upsilon_0 := \overline{\sigma}^{-1/2}\tau\,\overline{\sigma}^{-1/2}$, then it can be shown (see Theorem 4.3 in [2]) that the recursions

$$\upsilon_{2n+1}^{-1} = I + \gamma'\,\upsilon_{2n}\,\gamma \quad \text{and} \quad \upsilon_{2(n+1)}^{-1} = I + \gamma\,\upsilon_{2n+1}\,\gamma', \tag{A.2}$$

satisfy the Riccati difference equations

$$\upsilon_{2(n+1)} = \mathrm{Ricc}_\varpi\,(\upsilon_{2n}) \quad \text{and} \quad \upsilon_{2n+1} = \mathrm{Ricc}_{\overline{\varpi}}\,(\upsilon_{2n-1})\,, \quad \text{for } n \geqslant 0. \tag{A.3}$$

Let us also define

$$\tau_{2n} := \overline{\sigma}^{1/2}\ \upsilon_{2n}\ \overline{\sigma}^{1/2} \quad \text{and} \quad \tau_{2n+1} := \sigma^{1/2}\ \upsilon_{2n+1}\ \sigma^{1/2},$$

as well as the gain matrices

$$\beta_{2n} := \tau_{2n}\ \tau^{-1}\beta \quad \text{and} \quad \beta_{2n+1} := \tau_{2n+1}\ \beta'\tau^{-1}.$$

The recursions for τ_n and β_n are initialized with $(\beta_0, \tau_0) = (\beta, \tau)$, where (β, τ) are the parameters of the reference Gaussian measure.

Sinkhorn transitions. We also denote by G a d-dimensional centred Gaussian random variable (r.v.) with unit covariance. The Sinkhorn transitions in (2.2) can be expressed as

$$\mathcal{S}_{2n}(x, dy) = \mathbb{P}\left(\mathcal{Z}_{2n}(x) \in dy\right) \quad \text{and} \quad \mathcal{S}_{2n+1}(y, dx) = \mathbb{P}\left(\mathcal{Z}_{2n+1}(y) \in dx\right)$$

with random maps defined, for any $n \geqslant 0$, by

$$\begin{aligned} \mathcal{Z}_{2n}(x) &= m_{2n} + \beta_{2n}(x - m) + \tau_{2n}^{1/2}\ G \quad \text{and} \\ \mathcal{Z}_{2n+1}(y) &= m_{2n+1} + \beta_{2n+1}(y - \overline{m}) + \tau_{2n+1}^{1/2}\ G. \end{aligned}$$

The parameters m_n in the latter expressions are defined, for any $n \geqslant 0$, by the recursion

$$\begin{aligned} m_{2n+1} &= m + \beta_{2n+1}(\overline{m} - m_{2n}), \\ m_{2(n+1)} &= \overline{m} + \beta_{2(n+1)}\ (m - m_{2n+1}) \end{aligned}$$

with initial condition $m_0 = \alpha_0 + \beta_0 m$, where, again, $(\alpha_0, \beta_0) = (\alpha, \beta)$ are parameters of the reference Gaussian measure. Note that $\pi_n = \nu_{m_n, \sigma_n}$ with

$$\sigma_{2n} := \beta_{2n}\ \sigma\ \beta'_{2n} + \tau_{2n} \quad \text{and} \quad \sigma_{2n+1} := \beta_{2n+1}\ \overline{\sigma}\ \beta'_{2n+1} + \tau_{2n+1}.$$

Gibbs-loop process. The transitions of the Gibbs-loop process introduced in Section 2.2 can also be rewritten as

$$\mathcal{S}^{\circ}_{2n}(y_1, dy_2) := \mathbb{P}\left(\mathcal{Z}^{\circ}_{2n}(y_1) \in dy_2\right) \quad \text{and} \quad \mathcal{S}^{\circ}_{2n+1}(x_1, dx_2) := \mathbb{P}\left(\mathcal{Z}^{\circ}_{2n+1}(x_1) \in dx_2\right)$$

with the random maps

$$\begin{aligned} \mathcal{Z}^{\circ}_{2n}(y) &:= \overline{m} + \beta^{\circ}_{2n}(y - \overline{m}) + \left(\tau^{\circ}_{2n}\right)^{1/2}\ G \\ \mathcal{Z}^{\circ}_{2n+1}(x) &:= m + \beta^{\circ}_{2n+1}(x - m) + \left(\tau^{\circ}_{2n+1}\right)^{1/2}\ G \end{aligned}$$

defined in terms of the parameters

$$\begin{aligned} \beta^{\circ}_{2n+1} &:= \beta_{2n+1}\ \beta_{2n}, & \tau^{\circ}_{2n+1} &:= \tau_{2n+1} + \beta_{2n+1}\ \tau_{2n}\ \beta'_{2n+1}, \\ \beta^{\circ}_{2n} &:= \beta_{2n}\ \beta_{2n-1}, & \text{and } \tau^{\circ}_{2n} &:= \tau_{2n} + \beta_{2n}\ \tau_{2n-1}\ \beta'_{2n}. \end{aligned}$$

By Lemma 4.9 in [2], we have

$$0 \leqslant \overline{\sigma}^{-1/2}\ \beta^{\circ}_{2(n+1)}\ \overline{\sigma}^{1/2} = I - \upsilon_{2(n+1)} \leqslant (I + \varpi)^{-1}$$

as well as

$$0 \leqslant \sigma^{-1/2}\ \beta^{\circ}_{2n+1}\ \sigma^{1/2} = I - \upsilon_{2n+1} \leqslant (I + \overline{\varpi})^{-1}.$$

Moreover, Corollary 4.4 in [2] yields the uniform estimates

$$\overline{\sigma}^{1/2}(I + \varpi^{-1})^{-1}\overline{\sigma}^{1/2} \leqslant \tau_{2n} \leqslant \overline{\sigma} \quad \text{and} \quad \sigma^{1/2}(I + \overline{\varpi}^{-1})^{-1}\sigma^{1/2} \leqslant \tau_{2n+1} \leqslant \sigma,\ \forall n \geqslant 1.$$